

University of Colorado at Colorado Springs
Department of Mechanical and Aerospace Engineering

MAE 4425

HOMEWORK #6

DUE: Wednesday 5 August 2026

Solution by L. George 7/18/2026

Instructions:

- Include your references at the end of your homework in AIAA format.
<https://www.aiaa.org/publications/journals/reference-style-and-format>
- Include units in your final answers!
- Write non-mathematical answers in complete sentences.
- Clearly state your assumptions.

1. For **each** the current events presentations this week

- a) Summarize the presentation
- b) Describe something you learned from it
- c) Write one question you have left about the presentation

2. Carbon-carbon is a unique composite material consisting of carbon fibers embedded in a carbonaceous matrix. It was originally developed for aerospace applications; its low density, high thermal conductivity, and excellent mechanical properties at elevated temperatures make it an ideal material for aircraft brakes, rocket nozzles, and re-entry nose tips. It withstands temperatures in excess of 2000°C without major deformation.

In this problem, you are going to estimate the approximate number of photons per second that the Earth receives from the Sun, per unit area, which have sufficient energy to sever a single C-C bond. You may assume the bond energy is 25 eV = 3.47 eV.

a. Identify the maximum wavelength a photon may have and still be capable of severing the bond.

$$E = \frac{hc}{\lambda}$$

Where:

E = energy of a single photon (J)

h = Planck's constant = $6.62607015 \times 10^{-34}$ J-s

c = speed of light = 3×10^8 m/s

λ = wavelength (m)

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The bond energy of a single Carbon-Carbon bond is 3.47 eV and 1 eV = 1.60217663 x 10⁻¹⁹ J

$$\lambda = \frac{hc}{E}$$

$$\lambda = \frac{(6.62607015 \times 10^{-34} \text{ Js})(3 \times 10^8 \frac{m}{s})}{3.47 \times 1.60217663 \times 10^{-19}}$$

$$\lambda = 0.358 \mu m$$

b. Make a linear approximate for the solar irradiance $S(\lambda)$ over the waveband of interest. You may estimate from Figure 1.4 in the textbook or find your own source (be sure to reference it).

Estimated from Figure 1.4

$$S(\lambda) \text{ in } \frac{W}{cm^2 \mu m}$$

Wavelength (μm)	Irradiance $\frac{W}{cm^2 \mu m}$
0.2	0
0.4	0.19

$$\text{estimated slope} = \frac{0.19}{0.4 - 0.2} = 0.95$$

$$S(\lambda) = 0.95\lambda - 0.19 \frac{W}{cm^2 \mu m}$$

c. Integrate the ratio $\frac{S(\lambda)}{\text{photon energy}}$ over the waveband of interest to estimate the number of photons/s.

$$\frac{S(\lambda)}{\frac{hc}{\lambda}} = \frac{1}{hc} (0.95\lambda^2 - 0.19\lambda)$$



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$$\int_{0.2}^{0.36} \frac{1}{hc} (0.95\lambda^2 - 0.19\lambda) d\lambda$$

$$\text{Photons}/(sm^2) = 1.557 \times 10^{20}$$

- see Matlab code

Some students will work this out by hand and get confused on the units. Here is the “by-hand” solution.

$$\frac{S(\lambda)}{\frac{hc}{\lambda}} = \frac{1}{hc} (0.95\lambda^2 - 0.19\lambda) \frac{W\mu m}{cm^2 \mu m}$$

$$\int_{0.1}^{0.36} \frac{S(\lambda)}{\frac{hc}{\lambda}} = \int_{0.1}^{0.36} \frac{1}{hc} (0.95\lambda^2 - 0.19\lambda) d\lambda \frac{W\mu m^2}{cm^2 \mu m}$$

$$\int_{0.1}^{0.36} \frac{S(\lambda)}{\frac{hc}{\lambda}} = \frac{1}{hc} \left\{ \left(\frac{0.95(.36)^3}{3} - \frac{0.19(.36)^2}{2} \right) - \left(\frac{0.95(.1)^3}{3} - \frac{0.19(.1)^2}{2} \right) \right\} \frac{W\mu m}{cm^2}$$

$$\int_{0.1}^{0.36} \frac{S(\lambda)}{\frac{hc}{\lambda}} = \frac{1}{hc} (0.0024624 - (-.0006333)) \frac{W\mu m}{cm^2}$$

$$\int_{0.1}^{0.36} \frac{S(\lambda)}{\frac{hc}{\lambda}} = .0030957 \frac{W\mu m}{cm^2} * \frac{(100cm)^2}{m^2} * \frac{m}{10^6 \mu m}$$

$$\int_{0.1}^{0.36} \frac{S(\lambda)}{\frac{hc}{\lambda}} = \frac{3.0957 \times 10^{-5} \frac{W}{m} * \frac{J}{Ws}}{6.626 \times 10^{-34} Js * \left(3 \times 10^8 \frac{m}{s} \right)}$$

$$\frac{\text{Photons}}{m^2 s} = 1.557 \times 10^{20}$$

d. Approximately what percentage of photons from the sun is this? Note: This is only one method. It is the estimated total photon output based on a blackbody approximation. In question 2, you will use a more exact method.

i. Find the average photon energy (hint use the maximum wavelength output from the sun):

$$E = \frac{hc}{\lambda}$$

$$E = \frac{(6.626 \times 10^{-34} J \cdot s) * (3 \times 10^8 m/s)}{0.5 \times 10^{-6} m}$$

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$$E = 3.9756 \times 10^{-19} \text{ J/photon}$$

- ii. Assume a solar luminosity of 3.828×10^{26} Watts to find the number of photons per second

$$\text{Photons/sec} = \frac{3.828 \times 10^{26} \text{ J/s}}{3.9756 \times 10^{-19} \frac{\text{J}}{\text{photon}}}$$

$$\text{Photons/second} \approx 9.63 \times 10^{44}$$

- iii. Approximately what percentage of photons from the sun is this? Note units from the blackbody estimate are in Photons/second, whereas the integral results in units of photons/m²s. So to convert to proper units, take your answer from part ii and divide by $4\pi r^2$ where r is the distance to the Earth.

$$\frac{\text{Photons}}{\text{m}^2\text{s}} = \frac{9.63 \times 10^{44}}{4\pi(1 \text{ AU})^2} = \frac{9.63 \times 10^{44}}{4\pi(1.495979 \times 10^{11} \text{ m})^2}$$

$$\frac{\text{Photons}}{\text{m}^2\text{s}} = 3.4242 \times 10^{21}$$

$$\% = \frac{1.557 \times 10^{20}}{3.42425 \times 10^{21}}$$

4.5%

- e. Do you think this is this a significant risk for space applications?

Yes, although it's a low percentage, especially with a with a five or ten year mission it could be significant enough to weaken carbon-carbon composites. Recommend another composite or lining with beta-cloth liners to protect against UV radiation.

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3. Using the same assumptions you made in problem 1, but using Planck's law of black body radiation from the sun:

$$I_{\nu}(T) d\nu = \frac{2}{c^2} \frac{h\nu^3}{e^{h\nu/kT} - 1} d\nu$$

Planck
Function

$$I_{\lambda}(T) d\lambda = \frac{2hc^2 / \lambda^5}{\exp(hc / \lambda kT) - 1} d\lambda$$

$I_{\nu}(T)$ is the **specific intensity**, depends on T and ν

T is temp in Kelvin (Kelvin = Celsius + 273.2)

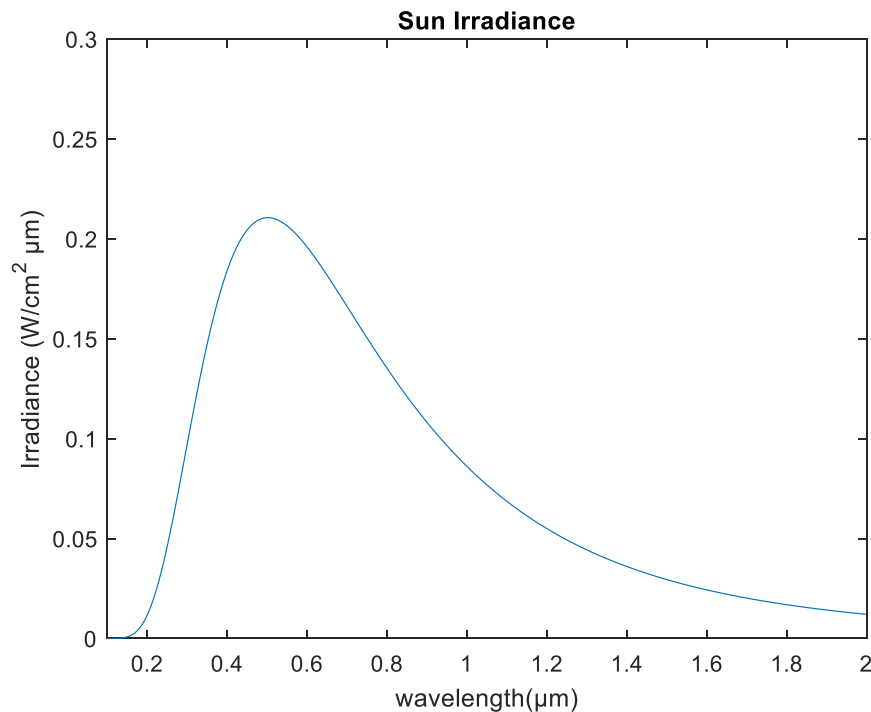
h is Planck's constant: 6.636×10^{-34} J s

k is Boltzmann's constant 1.38×10^{-23} m² kg s⁻² K⁻¹

c is the speed of light

a. Noting that the law applies to a perfect blackbody, use it to approximate figure 1.4 for $S(\lambda)$ in the textbook.

See Matlab code



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b. Integrate the ratio $[S(\lambda)/\text{photon energy}]$ over the waveband of interest to re-estimate your solution to problem 1.

Problem 1: $\text{Photons}/(m^2s) = 1.557 \times 10^{20}$

Problem 2: $\text{Photons}/(m^2s) = 1.5958 \times 10^{20}$

c. How closely do your two answers agree?

Within 2.4%

d. Now estimate the total number of photons/ (m^2s) that the earth receives from the sun from your results in part a. What percentage of your answer in b does this represent?

Now integrate over entire curve (see Matlab code)

$1.8875 \times 10^{21} \text{ photons}/(m^2s)$

$$\% = \frac{1.5958 \times 10^{20}}{1.8875 \times 10^{21}}$$

8.45%

e. How closely does your answer here compare to the estimate you used in problem 1 d.

Answer 1d: 4.5%

Answer 2d: 8.45%

46% difference

f. Which answer do you believe is more accurate and why?

The method in problem 2. However, both were estimated based on major assumptions (a linear solar irradiance in problem 1 versus a black body in problem 2) so neither is entirely accurate.

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4. NASA is designing an interplanetary probe to investigate Eris. Once the probe arrives at the dwarf planet, only a low-resolution imager along with other instrumentation that can function at very low temperatures can be used. To maximize the value of the program, the mission manager has been asked to consider which planets along the way the spacecraft could take high-resolution images of. The high-resolution camera must be kept between -35°C to 35°C . Find the equilibrium temperature of the probe at each planet and the dwarf planet Pluto. You must consider solar input, albedo, and planetary infrared effects.

You may make the following assumptions. Clearly state any additional assumptions you make.

- You may not make any modifications to the spacecraft
 - The probe will be in orbit around each planet long enough to reach equilibrium temperature.
 - The probe is cube shaped with each side 1 m with an absorptivity, α , or 0.3 and emissivity, ϵ , or 0.7
 - The probe internal components generate 750W of heat
 - One side always faces the sun
 - All sides emit radiative energy at the same rate
- a. Plot the equilibrium temperatures for each of the planets.

Solution:

$$Q_{solar} + Q_{albedo} + Q_{IR} + Q_{internal} = Q_{emitted} = \epsilon \sigma A T^4$$

Example for Mars:

$$Q_{solar} = \alpha * Area * Solar\ Flux$$

$$Solar\ Flux = Solar\ Flux_{earth} \left(\frac{au}{r}\right)^2$$
$$Solar\ Flux = \frac{1366.1\ W}{1.524^2\ m^2} = 588.2\ \frac{W}{m^2}$$

$$Q_{solar} = 0.3 * 1m^2 * 588.2\ \frac{W}{m^2}$$
$$Q_{solar} = 176.5\ W$$

$$Q_{albedo} = \alpha * Area * Albedo\ Flux$$
$$Albedo\ Flux = 0.29 * 588.2\ \frac{W}{m^2} = 170.6\ \frac{W}{m^2}$$
$$Q_{albedo} = 0.3 * 1m^2 * 170.6\ \frac{W}{m^2}$$
$$Q_{albedo} = 51.2\ W$$

$$Q_{IR} = 0.3 * 1m^2 * 141\ \frac{W}{m^2}$$

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(based on average IR flux)

$$Q_{IR} = 42.3 \text{ W}$$

$$Q_{internal} = 750 \text{ W}$$

$$Q_{solar} + Q_{albedo} + Q_{IR} + Q_{internal} = Q_{emitted} = \epsilon \sigma A T^4$$

$$(176.5 + 51.2 + 42.3 + 750) \text{ W} = 0.7 * (5.670374419 \text{E}^{-8} \frac{\text{W}}{\text{m}^2 \text{K}^4}) * 6 * T^4$$

Equilibrium temperature = 255.8K

Equilibrium temperature = -17.3 °C

See Matlab Code for remaining results in °C

Mars = -17.3

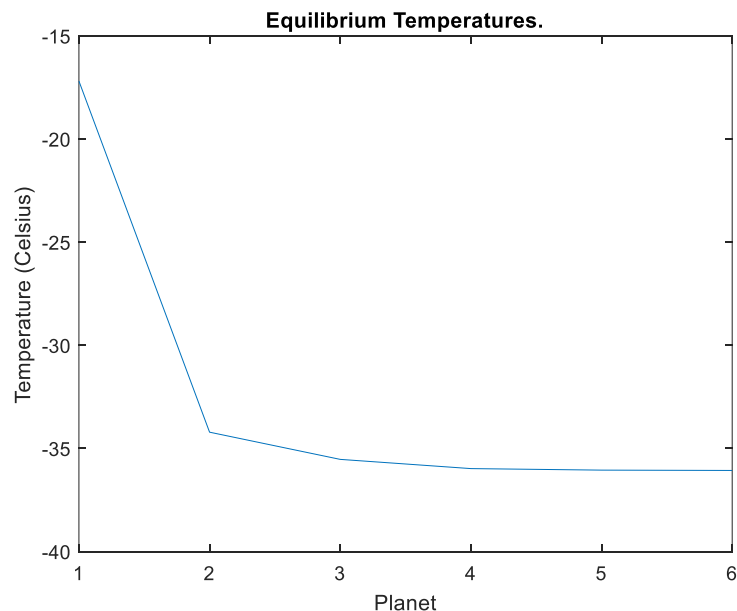
Jupiter = -34.2

Saturn = -35.5

Uranus = -36.0

Neptune = -36.05

Pluto = -36.06



b. Recommend, from a thermal control perspective, which planets to image.

Mars and Jupiter are good additional targets for this mission.

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5. You have been given a budget of \$15K to design a thermal control system for the mission and a goal of imaging as many planets as possible along the way. You have the following choices (or you may research your own methods). Assume each kg of mass added adds \$25,000 to the cost of the mission.

Thermal Coating	Emissivity	Mass	Notes
Insulation	0.05	0.3 kg/m ²	Passive system
White paint*	0.85	Negligible	Passive system
Black paint**	0.874	Negligible	Passive system
Radiators	0.8	0.6 kg/m ²	Passive system
Radiations w/louvers°	0.05 – 0.8	2.1 kg/m ²	Active system
Heaters ^Δ	-----	0.025 kg/m ²	Active system

*Absorptivity of 0.252

**Absorptivity of 0.975

° Each location requires the use of a controller with a mass of .2 kg and power requirement of 2 W. Also indicate range of emissivity used in your design

^Δ Thermal input based on power requirements

Adding insulation to 2 sides of the spacecraft yields (for Pluto – worst case, power of internal components)

$$750 \text{ W} = (2 \cdot .05 + 4 \cdot .7) \cdot (5.670374419 \text{E}^{-8} \frac{\text{W}}{\text{m}^2 \text{K}^4}) \cdot T^4$$

Yields

$$T = 259 \text{ K} = -13^\circ \text{ C}$$

This will be at a cost of .3 kg x 2 = .6 kg

Or \$15,000

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6. An outgassing test of Neoprene showed that the outgassing rate is $10^{-5} \frac{\text{Torr liter}}{\text{cm}^2 \text{s}}$ at a temperature of 298K.

a. Show that the outgassing rate in $\frac{W}{m^2}$ can be expressed as $7.5 \times 10^{-4} \frac{\text{Torr liter}}{\text{cm}^2 \text{s}}$

Noting that:

$$1 \text{ Torr} = 133.322368 \text{ Pa}$$

$$\text{Pa} = 1 \frac{\text{N}}{\text{m}^2}$$

$$1 \text{ liter} = 0.001 \text{ m}^3$$

$$W = \frac{J}{s}$$

$$J = \text{N m}$$

$$\begin{aligned} 1 \frac{W}{m^2} &= 1 \frac{J}{s m^2} = 1 \frac{N}{s m} = 1 \frac{\text{Pa m}}{s} \\ 1 \frac{\text{Pa m}}{s} &\times 1 \frac{\text{Torr}}{133.322368 \text{ Pa}} = \frac{1}{133.322368} \frac{\text{Torr m}}{s} \\ &= \frac{1}{133.322368} \frac{\text{Torr m}}{s} \times \frac{\text{liter}}{0.001 \text{ m}^3} \times \frac{\text{m}^2}{(100 \text{ cm})^2} \end{aligned}$$

So

$$1 \frac{W}{m^2} = 7.5 \times 10^{-4} \frac{\text{Torr liter}}{\text{cm}^2 \text{s}}$$

b. Determine the number of molecules released per unit area per second in $\frac{\text{molecules}}{\text{cm}^2 \text{s}}$

Since

$$q(t) = k_B T \dot{n}(t)$$

$$\dot{n}(t) = \frac{q(t)}{k_B T}$$

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$$\dot{n}(t) = \frac{q(t)}{k_B T} = \frac{10^{-5} \frac{\text{Torr liter}}{\text{cm}^2 \text{s}} \times \frac{1 \frac{\text{W}}{\text{m}^2}}{7.5 \times 10^{-4} \frac{\text{Torr liter}}{\text{cm}^2 \text{s}}}}{1.3806452 \times 10^{-23} \frac{\text{J}}{\text{K}} \times 298 \text{K}}$$

$$\dot{n}(t) = \frac{q(t)}{k_B T} = 3.2407 \times 10^{18} \frac{\frac{\text{W}}{\text{m}^2}}{\frac{\text{J}}{\text{K}}} \times \frac{\frac{\text{J}}{\text{s}}}{\text{W}} \times \frac{\text{m}^2}{(100 \text{cm})^2}$$

$$\dot{n}(t) = 3.2407 \times 10^{14} \frac{\text{molecules}}{\text{cm}^2 \text{s}}$$

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7. Classification of air cleanliness in cleanrooms is specified by ISO-14644-1. The classifications are defined in terms of the concentration of airborne particulate contamination to levels appropriate for accomplishing contamination sensitive activities. Using the equation from Lesson 3 for the maximum number of particles permitted for any given size:

$$\frac{\text{Particles}}{m^3} = 10^M \left(\frac{0.1 \mu m}{d} \right)^{2.08}$$

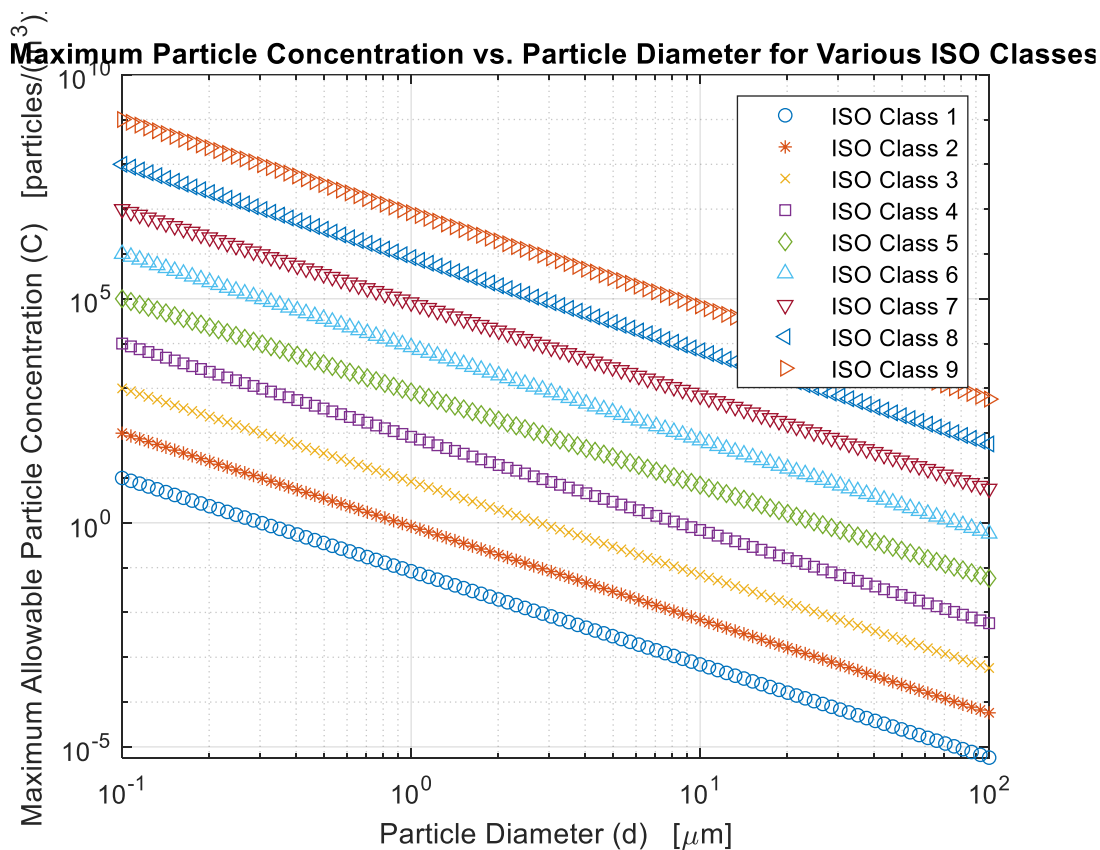
Where:

M = numerical designation of the class (SI units)

d (μm) = particle size

Create a plot of the maximum cleanroom airport particle concentration for each ISO classification. Your plot should look something like Fig 2.13 in the textbook but will be slightly different since the textbook figure is based on the old FED STD 209E. Hint: use log-log axes.

[See Matlab code](#)



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8. An ASTM E-595-07 test showed that a 10 m x 10 cm test specimen of Kapton 1 mil (0.001 inch) thick had a TML of 0.5%, during which only the top side was exposed. Determine the outgassing rate in units of $\frac{\text{Torr-liter}}{\text{cm}^2\text{s}}$. The mass density is 1.4 g/cm³ and the molar mass of the outgassed products is 15 g/mol. Clearly state your assumptions.

$$q = \frac{\dot{m} N_A k_B T}{M}$$

Start by finding $\dot{m}(t)$ = mass loss per unit area per unit time, $\frac{\text{kg}}{\text{m}^2\text{s}}$.

First find total mass = ρV .

$$\text{mass} = \frac{1.4\text{g}}{\text{cm}^3} (10\text{m}) \left(\frac{100\text{cm}}{\text{m}} \right) (10\text{cm}) (0.001\text{in}) \left(\frac{2.54\text{cm}}{\text{in}} \right) = 35.56\text{ g}$$

Now using TML = 0.5%

$$0.005 = \frac{\text{mass loss g}}{35.56\text{ g}}$$

$$\text{Mass loss g} = 0.1778\text{ g}$$

$$\text{So } \dot{m} = \frac{0.1778\text{g}}{24\text{ hrs}} \left(\frac{\text{hr}}{3600\text{s}} \right) = 2.0579 \times 10^{-6} \frac{\text{g}}{\text{s}}$$

$$N_A = 6.0221367 \times 10^{26} \frac{\text{particles}}{\text{kmol}}$$

$$k_B = 1.380649 \times 10^{-23} \frac{\text{J}}{\text{K}}$$

$$T = 398.15\text{ K (assume at } 125^\circ\text{C)}$$

$$M = 15\text{ g/mol}$$

$$q(t) = \frac{(2.05787 \times 10^{-6} \frac{\text{g}}{\text{s}}) (6.0221367 \times 10^{26} \frac{\text{particles}}{\text{kmol}}) (1.380649 \times 10^{-23} \frac{\text{J}}{\text{K}}) (398.15\text{ K})}{(15 \frac{\text{g}}{\text{mol}}) (1000 \frac{\text{mol}}{\text{kmol}})}$$

$$q(t) = 5.4516 \times 10^{-4} \frac{\text{W}}{\text{m}^2}$$

But from problem 4a, we know

$$\frac{\text{W}}{\text{m}^2} = 7.5 \times 10^{-4} \frac{\text{Torr liter}}{\text{cm}^2\text{s}}$$

$$q = 3.41 \times 10^{-7} \frac{\text{Torr liter}}{\text{cm}^2\text{s}}$$

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9. Research an innovative new heat transfer method used on spacecraft. Summarize it and the benefits provided by the new technology. Prepare a short summary to be presented to class.

Note: you may choose how you want to present it to class. Some options are below, or you may propose your own.

- classroom presentation
- handout on the technology
- video
- discussion board